

Frontend: Theory Assignment 2

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Q1. What is CSS Box Model?

CSS Box Model is used to describe how an HTML element takes space on a webpage. Every element is treated as a box.

It has four main parts: **Content** (text or image), **Padding** (space around content), **Border** (line around the element), and **Margin** (space outside the border).

Code:

```
div {
  width: 200px;
  padding: 10px;
  border: 2px solid black;
  margin: 20px;
}
```

Q2. What is media query? Explain with a sample code.

A media query is used to make a webpage responsive. It applies different CSS styles depending on the screen size or device.

For example, we can change the width and background color of a box when the screen becomes small.

Code:

```
<!DOCTYPE html>
<html>
<head>
<style>
.box {
  width: 500px;
  background-color: lightblue;
  padding: 20px;
}

@media (max-width: 600px) {
  .box {
    width: 100%;
    background-color: lightgreen;
  }
}
</style>
</head>
<body>

<div class="box">
  This is a responsive box.
</div>

</body>
</html>
```

Here, when the screen width is 600px or less, the media query style is applied.

Q3. What are the features of Bootstrap?

Bootstrap is a popular CSS framework used to create webpages easily. Some of its main features are:

- Responsive grid system
- Ready-made buttons, forms, cards and navigation bars
- Mobile-friendly design
- Predefined CSS classes
- Faster and easier webpage development
- Support for HTML, CSS and JavaScript

Q4. Write an example of basic grid structure in Bootstrap.

Bootstrap uses a 12-column grid system. A row is created using the **.row** class and columns are created using classes such as **.col-md-4**.

Code:

```
<div class="container">
  <div class="row">

    <div class="col-md-4">
      Column 1
    </div>

    <div class="col-md-4">
      Column 2
    </div>

    <div class="col-md-4">
      Column 3
    </div>

  </div>
</div>
```

Here, $4 + 4 + 4 = 12$, so three equal columns are created.

Q5. What is CSS Selectors?

CSS selectors are used to select HTML elements on which we want to apply CSS. There are different types of selectors:

- **Element selector** – selects an HTML tag.
- **Class selector** – selects elements having a particular class.
- **ID selector** – selects an element using its ID.
- **Universal selector** – selects all elements.
- **Group selector** – selects multiple elements together.

Code:

```
p {
  color: blue;
}

.box {
  background-color: yellow;
}

#heading {
  font-size: 25px;
}
```

Q6. What are CSS backgrounds? List the properties.

CSS background properties are used to set the background of an HTML element. We can use a color, image and other settings for the background.

Main properties are: **background-color**, **background-image**, **background-repeat**, **background-position**, **background-size**, and **background-attachment**.

Code:

```
body {
```

```

background-color: lightblue;
background-image: url("image.jpg");
background-size: cover;
background-position: center;
}

```

Q7. How can we overlap elements in CSS?

Elements can be overlapped using CSS positioning and the **z-index** property. Usually, **position: absolute** is used to place one element over another.

The **z-index** property decides which element appears in front.

Code:

```

.box1 {
  width: 150px;
  height: 150px;
  background-color: lightblue;
  position: absolute;
}

.box2 {
  width: 150px;
  height: 150px;
  background-color: lightcoral;
  position: absolute;
  top: 50px;
  left: 50px;
  z-index: 2;
}

```

Here, Box 2 overlaps Box 1 and appears above it because its z-index is higher.

Q8. What are the different positioning properties in CSS?

The CSS **position** property decides how an element is placed on a webpage.

- **static** – default position of an element.
- **relative** – moves an element from its normal position.
- **absolute** – positions an element relative to its positioned parent.
- **fixed** – keeps an element fixed on the screen while scrolling.
- **sticky** – stays in its normal position until a scroll point is reached, then sticks.

Code:

```

.box {
  position: relative;
  top: 20px;
  left: 30px;
}

```

Here, the element moves 20px down and 30px to the right from its normal position.

Q9. What is ECMAScript?

ECMAScript is a standard or specification for scripting languages. JavaScript follows the ECMAScript standard.

It defines rules for features such as variables, functions, objects and data types.

ECMAScript is updated from time to time with new features. For example, ES6 introduced **let**, **const**, arrow functions and classes.

So, JavaScript is an implementation of the ECMAScript standard.

Q10. What is the difference between var, let and const in JavaScript? Explain Temporal Dead Zone.

var, let and const are used to declare variables in JavaScript.

Feature	var	let	const
Scope	Function	Block	Block
Re-declaration	Allowed	Not allowed	Not allowed
Re-assignment	Allowed	Allowed	Not allowed
Hoisting	Yes	Yes, TDZ applies	Yes, TDZ applies

- **var** can be declared again in the same scope.
- **let** can be changed but cannot be declared again in the same block.
- **const** cannot be reassigned after its value is given.
- let and const are block scoped, while var is function scoped.

Example:

```
var a = 10;
a = 20;

let b = 30;
b = 40;

const c = 50;
// c = 60; // Error
```

Temporal Dead Zone (TDZ): TDZ is the time between entering a block and the actual declaration of a let or const variable. During this time, the variable cannot be accessed.

```
console.log(x); // Error

let x = 10;
```

Here, x exists in the scope but cannot be used before its declaration. This period is called the Temporal Dead Zone.